

constantan wires.

Thermocouples are very popular because of their low thermal mass and wide operating temperature range, which can extend to about 1700°C with common types. However, sensitivity of thermocouples is rather small (of the order of tens of microvolts per°C). A low-offset amplifier is needed to produce a usable output voltage.

Thermistors

Thermistors are special solid temperature sensors that behave like temperature-sensitive electrical resistors. These are generally composed of semiconductor materials. There are basically two types of thermistors—negative temperature coefficient (NTC), which are used mostly in temperature sensing and positive temperature coefficient (PTC), which are used mostly in electric current control.

Thermistor exhibits a change in electrical resistance with a change in its temperature. The resistance is measured by passing a small, measured direct current through it and measuring the voltage drop produced thereby. When it comes to NTC-type, the negative coefficient can be as large as several per cent per°C, allowing the thermistor circuit to detect minute changes in temperature, which could not be observed with a thermocouple circuit.

Low-cost thermistors often perform simple measurement (and trip-point detection) functions in low-end systems. Low-precision thermistors are often inexpensive. You can find thermistors that will work over a temperature range from about -100°C to +550°C, although most are rated for maximum operating temperatures from 100°C to 150°C. Simple thermistor-based set-point thermostat or controller applications can be implemented with very few components. Just the thermistor, a comparator and a few other components can do the job.

As thermistors are extremely non-linear devices that are highly dependent upon process parameters, and their performance may be degraded by self-



PTC thermistors

heating, these have drawbacks in some applications. For example, resistance temperature function of a thermistor is very non-linear, so if wide range of temperatures are to be measured, you'll find it necessary to perform substantial linearisation.

Sensor ICs

There are a wide variety of temperature sensor ICs that are available to simplify the broadest possible range of temperature monitoring challenges. These silicon temperature sensors differ significantly from the above mentioned types in a couple of important ways.

The first is operating temperature range. A temperature sensor IC can operate over the nominal IC temperature range of -55°C to +150°C. The second major difference is functionality. A silicon temperature sensor is an integrated circuit, and can therefore include extensive signal processing circuitry within the same package as the sensor. There is no need to add compensation (or linearisation) circuits for temperature sensor ICs.

Some of these are analogue circuits with either voltage or current output. Others combine analogue-sensing circuits with voltage comparators to provide alert functions. Some other sensor ICs combine analogue-sensing circuitry with digital input/output and control registers, making them an ideal solution for microprocessor-based systems.

Digital output sensor usually contains a temperature sensor, an

analogue-to-digital converter (ADC), a two-wire digital interface and registers for controlling the IC's operation. Temperature is continuously measured and can be read at any time. If desired, the host processor can instruct the sensor to monitor temperature and take an output pin high (or low) if temperature exceeds a programmed limit. Lower threshold temperature can also be programmed and the host can be notified when temperature has dropped below this threshold. Thus, digital output sensor can be used for reliable temperature monitoring in microprocessor-based systems.

How to use?

A temperature sensor produces an analogue or digital output whose strength depends on the temperature of the sensor. Heat is conducted to the sensing element through the sensor's package and its metal leads. In general, a sensor in a metal package will have a dominant thermal path through the package. For sensors in plastic packages, the leads provide the dominant thermal path. Therefore a board-mounted IC sensor will do a fine job of measuring the temperature of the circuit board.

If it's needed to measure the temperature of something other than the circuit board, it should be ensured that the sensor and its leads are at the same temperature as the object you wish to measure. This usually involves making a good mechanical (and thermal) contact by attaching the sensor and its leads to the object being measured with thermally-conductive epoxy.

If a liquid's temperature is to be measured, the sensor can be mounted inside a sealed-end metal tube and dipped into a bath, or screwed into a threaded hole in a tank. Temperature sensors and any accompanying wiring and circuits must be kept insulated and dry, to avoid leakage and corrosion.

Any linear circuit connected to wires in an uncongenial environment can have its performance adversely affected by intense electromagnetic